Optimization for Deep Learning (OPT4DL)

Abolfazl Hashemi, Purdue ECE

Optimization Basics

Stationary Points



- Consider

$$\min_{w} f(w) \qquad s.t. \quad w \in \mathbb{R}^d \tag{1}$$

- w^* is a stationary point if the gradient (derivative) of f at w^* is zero:

$$\nabla f(w^*) = 0 \tag{2}$$

- If further $\nabla^2 f(w^*) \succ 0$, then w^* is a local minimum
- If further $0 \succ \nabla^2 f(w^*)$, then w^* is a local maximum
- A stationary (or even a local minimum) is not necessarily an optimal solution

Convex Sets



- A set $\mathcal{X}$ is convex if for all x_i in $\mathcal{X}$ and all $m \geq 1$

$$\sum_{i=1}^{m} w_i x_i \in \mathcal{X} \tag{3}$$

where

$$w_i \geq 0, \quad \sum_{i=1}^{m} w_i = 1 \tag{4}$$

- In other words, the convex combination of all points belong to the set as well.

Convex Functions



- f is (strongly) convex if for all x_i

$$f(\sum_{i=1}^{m} w_i x_i) \leq \sum_{i=1}^{m} w_i f(x_i) \tag{5}$$

where

$$w_i \geq 0, \quad \sum_{i=1}^{m} w_i = 1 \tag{6}$$

($<$ for strongly convex case instead of $\leq$)

Convex Functions (Cont'd)



- f is (strongly) convex if all x and y

$$f(\mathrm{y}) \geq f(x) + \langle \nabla f(x), \mathrm{y} - x \rangle + \frac{\mu}{2} \|\mathrm{y} - x\|^2 \tag{7}$$

Convex Functions (Cont'd)



- f is (strongly) convex if for all x

$$\nabla^2 f(x) \succeq 0 \tag{8}$$

($\succ$ for strongly convex case instead of $\succeq$)

Why Convexity is Nice?



- Local information provides global information
- Any stationary point w^* (can be identified locally) is a global minimum (a property requiring global information about the function in general)
- Proof is easy to see from the definition

$$f(\mathrm{y}) \geq f(w^*) + \langle \nabla f(w^*), \mathrm{y} - w^* \rangle \tag{9}$$

  and noting $\nabla f(w^*) = 0$
- Furthermore, A strongly convex function has a unique minimizer

Convergence Notions



- We are interested in establishing

$$\mathrm{err}(\tilde{w}) \leq \epsilon \tag{10}$$

  for some target accuracy ϵ and a notion of convergence $\mathrm{err}(\cdot)$
- Our methods achieving this goal are typically iterative, i.e., they find an approximate solution in T iterations
- Typically better accuracy (lower ϵ) requires larger T
  - Sublinear rates: $T = \Omega(\epsilon^{-a})$, for some $a > 0$
  - Linear rates: $T = \Omega(\log \frac{1}{\epsilon})$
- Typical examples of $\mathrm{err}(\tilde{w})$
  - approximate first-order: $\|\nabla f(\tilde{w})\| \leq \epsilon$, for some $a > 0$ (General functions)
  - Low suboptimality gap: $f(\tilde{w}) - f^* \leq \epsilon$ (Convex)
  - Low point-wise error $\|\tilde{w} - w^*\| \leq \epsilon$ (Strongly convex functions)

Optimization as a Fixed Point Problem



- Consider

$$\min_{w} f(w) \qquad s.t. \quad w \in \mathbb{R}^d \tag{11}$$

- A necessary condition for optimality is first-order stationarity $\nabla f(w^*) = 0$
- Equivalently

$$\eta \nabla f(w^*) - w^* = -w^* \tag{12}$$

- Equivalently, the problem is to find the fixed points of $g(w) := w - \eta \nabla f(w)$
- The simplest method is the Fixed Point Iteration $w_{t+1} = g(w_t)$, for all $t = 1, \ldots, T$ which leads to

$$w_{t+1} = w_t - \eta \nabla f(w_t) \tag{13}$$

- This method is called Gradient Descent (GD) with stepsize/learning rate $\eta > 0$ which serves as the foundational of all DL training methods.